

[VENKATESH GANESAN | LinkedIn](#)

Principal Automation Engineer | Hybrid-Cloud | Video Platform
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Professional Summary

- 15 years of experience as Principal Software Engineer extensively involved in developing, testing, implementing & managing scalable application in domains like **Engineering Design R&D, Video Engineering, Optical-Network & Cloud Computing**.
- Act as a technical mentor and handled all the ceremony of **scrum and kanban** as a technical lead, effectively lead cross-functional teams, ensuring clear communication among product owner, subject matter experts & stockholders.
- Handling incidents and issues related to software development, **automation, infrastructure**, troubleshooting problems, root cause analysis and implementing solutions to prevent future occurrences.
- Worked as a technical engineer in developing/testing it in lab (On-premises/Cloud) for **IPTV/VOD/TimeShift** system using **Wowza, Adobe, Microsoft Mediaroom, FFMPEG**.
- Expert knowledge in **Video codec and streaming media** and its implementation over IP (unicast/multicast).
- Worked on development of automation system, traffic generation and its monitoring system for video delivery.

Technical Skills

Language & Framework	Core Java, Python, Flask, SQLAlchemy, PyMongo, PyUnit, Py Test
Cloud Computing	AWS, OpenStack, VMwareExsi, AWS- Lambda, AWSSDKFORJAVA, AWSSDK-BOTO3, Cloud Formation, Terraform, EC2, S3, Auto scaling, Trusted Advisor, VPC, Direct Connect, Route 53, Cloud watch, Cloud trail, IAM & Roles.
OS & Containers	Windows, Android, iOS, Linux, JujuCharm, Kubernetes.
DevOps Tool & IDE	Git, Maven, Jenkins, Json, Yaml, Splunk, Bugzilla, Jira, Confluence, Nagios, Wireframe, Wireshark, Visual Studio Code, Eclipse, SQL Developer, Tortoise SVN, RestApi, Postman.
Application & Media Server	Ffmpeg, Wowza Media Server, Adobe Media Server, Microsoft Mediaroom, Live555, Nginx, Apache Tomcat.
Codec, Protocol & Streaming Media	Rtp, Rtsp, Rtmp, Http, Tcp, Udp, H.264, AVC, Mpeg-Dash, Hls, Hds, Cdn, Drm, Dvr, Encoding, Transcoding, Transmuxing, Cache, IPTV, Ott, Webrtc.

Work Experience

Ornge Inc - Cupertino, CA – System Architect (Oct 2023 – Present)

- Worked on proof of concept to automate deployment & scaling of Api using Juju Charm (Kubernetes) in Linux environment
- Mentor juniors to integrate Juju charms with CI/CD pipelines, ensuring smooth deployments and updates
- Provided technical support & troubleshooting charm issues, including debugging & resolving deployment failures
- Proficient in managing network, hardware, and storage systems to support seamless delivery of **live stream/vod/time shift streaming** content across multiple platforms using **wowza media server**.
- Monitor server health, network bandwidth, and storage utilization in real-time, using tools such as Zabbix, Nagios, and Prometheus.
- Collaborate with DevOps and Network Engineering teams to optimize the **streaming infrastructure**, enabling real-time content delivery with **minimal latency**.
- Utilize Content Delivery Networks (CDNs) to ensure fast and efficient **video streaming** to global audiences while minimizing buffering

Tech Mahindra – Tech Lead (Aug 2022 - Oct 2023)

- Improved live stream reliability by implementing automated failover systems spearheaded the implementation of cloud-native streaming platforms and optimizing CDN configurations and improving server-to-edge delivery paths.
- Implemented best practices for security and scalability while migrating from on Prem to cloud deployment.
- Worked closely with content acquisition and **video encoding** teams to ensure high-quality **live streams** were optimized for various device platforms (mobile, desktop, smart TVs)
- Assisted in the integration of cloud-based video streaming solutions, improving **scalability during peak live events**.
- Developed infrastructure-as-code scripts using AWS CloudFormation and Terraform to automate resource provisioning.
- Experience in operating and troubleshooting within a virtualized environment
- Experience with network architectures, technologies, intrusion detection technologies and security risk identification
- Experience with Linux programming and scripting (bash, python)
- Experience monitoring production systems, performance optimization, root cause analysis, and advanced troubleshooting
- Serves as the technical expert responsible for the engineering, implementation, installation, operation, and maintenance of data archive and backup within Linux Red Hat and Centos networks

Bridgera – Tech Lead (Mar 2022 - Jul 2022)

- Extensive knowledge of all Quality Center modules including Management, Requirements, Test Plan, Test Lab and Defects
- Provide test document which include test plan, test script (manual and/or automation), summary report & defect logging
- Experience in working in virtual environments and setting up virtual testbed environments
- Familiarity with CI/CD & configuration management and defect tracking tool (Jenkins, GIT, JIRA)
- Documented validation processes and created detailed reports for stakeholder's

Wowza Media Systems – Production Support Engineer (Oct 2021 - Jan 2022)

- Worked as a lead technical support engineer for streaming and achieve **low latency**.
- Experience providing 'on call' support on a rotating basis to resolve system/application issues
- Experience gathering and analyzing data to optimize system health/performance and acting as a point of escalation for **troubleshooting production issues for live streaming**
- Experience working with streaming protocols & CDN within an engineering capacity (Akamai, AWS CloudFront)

Wipro Limited, Senior Project Engineer –IPTV/Optical Network - Nokia & Comcast (Jan 2015 - Sept 2021)

- Worked on the development of automation system for **cdvr product(Fabrix)** and test & implemented in the lab.
- Hands-on experience in **end-to-end video delivery technologies, adaptive bitrate streaming** and content protection, optimized & maintained video streaming workflow for both **live & OnDemand streaming** with **Microsoft media room**.
- Automation script development to monitor, debug and efficiently operate the **video streaming infrastructure** using python
- Debugging potential video streaming issues across different platforms and protocols by providing technical solutions and extensive knowledge in **video workflow cdn and drm**
- Troubleshoot and resolved issues related to video playback and streaming quality for **Android, iOS and set top box devices**
- Developed and maintained video streaming applications and implemented adaptive bitrate streaming to ensure smooth playback across various network conditions
- Fetch data from different data sources in aws cloud environment (Catchpoint, Jira, Bugzilla and Splunk) & extract using boto3 and transform it to database and tableau dashboard using python
- Worked on the development of Automation framework for network planning tool (Optical Network Planning Tool - PSS1830)
- Development of test case for network (Engineering Planning Tool-OSI-Layer0) product and also worked on the development of the UI automation using Sikuli for Network Planning Tool.
- Experienced in defining, collecting, tracking system-level, hardware, & software requirements for provisioning & create a wireframe to provides an overview of network performance, usage, alerts and also generates reports on network usage, performance metrics, security incidents & automates network configuration and management based on predefined policies
- Developed a comprehensive monitoring tool using AWS SDK for Java to track and analyze cloud resources using Core Java and Maven for implementing business logic, backend services, project dependencies and build processes.
- Integrated AWS services such as EC2, CloudWatch, S3 and trusted advisor for real-time monitoring and data storage.

3JTech – Senior Software Engineer (IP-CAMERA - Feb 2014 - Dec 2014)

- Developed and maintained a multimedia application (**IP-CAM**) using FFmpeg, C++ & C#
- Worked on video encoding, decoding, transcoding & improved video quality and tested in firmware.
- Integrated FFmpeg libraries to support **h.264** video formats and codecs from **MPEG-4** ensuring high-quality video output
- Implemented real-time video streaming features, including **RTSP and HLS protocols**, for seamless video delivery
- Conducted thorough testing and debugging to ensure firmware stability and reliability
- Collaborated with hardware engineers to integrate firmware with camera hardware components

Firesky Media Technologies Inc- Software Developer (IPTV - Aug 2012 - Jan 2014)

- Developed live streaming application using **Wowza SDK** for Java enabling **real-time video broadcasting and playback**
- Implemented custom Java classes to extend media server functionalities and enhance user experience for **IPTV**
- Integrated hardware components such as encoder, tuners, decoders, and network interfaces conducted rigorous testing and debugging to ensure system stability and performance
- Experience with **IP multicast, MPEG-DASH, Rtsp, Rtmp, Apple HLS, Adobe HDS, Windows HSS, VOD, IPTV**, cloud encoding, transcoding, packaging live streaming, video codec, play back & performed troubleshooting of streaming video
- Worked on Logic Analyzers and also Unit Testing, Integration Testing, System Testing and regression.
- Worked on Interfaces like USB, Ethernet and development tools like Debuggers, Compilers, Emulators, Test Equipment
- Working knowledge of **IP networking, TCP/IP protocols, UDP, Http, Routing and Switching** and audio mixer
- Familiar with troubleshooting by using **Wireshark/TCP dump** or collecting/decoding traces/logs from applications
- Experience in designing, implementing and supporting **Encode and Decode** both HW/SW

Technosoft Inc – Senior Software Engineer (2D/3D - May 2008 - June 2011)

- Developing detailed 2D and 3D models and drawings using Adaptive Modeling Language-AML.
- Analyze specifications, sketches, and related data to assess factors affecting component designs.
- Modify and revise designs to correct deficiencies or reduce production problems
- Work closely with other engineering departments to ensure design intent is accurately translated into production
- Maintain documentation of all design drafts and revisions
- Create and maintain accurate BOMs, ensuring all components, materials, and sub-assemblies are correctly listed
- Conduct testing and validation of designs to ensure they meet specifications and industry standards
- Optimize designs for manufacturability, cost-effectiveness, and performance.